

According to the New Book & Syllabus of
Board of Intermediate, Sindh.

THE STUDENTS' COMPANION OF *Mathematics*

For Class **11**

INCLUDES **MODEL PAPER** FOR FURTHER PRACTICE



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Chapter # 03 VECTORS

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Chapter # 04 SEQUENCE & SERIES

CHAPTER # 03 VECTORS**EXERCISE 3.1****Q.1 Classify the following quantities as scalars or vectors:**

- (i) Force (ii) Speed (iii) Velocity (iv) Area
 (v) Acceleration (vi) Work (vii) Volume (viii) Weight
 (ix) Mass

SOLUTION:

(i) Force	(ii) Speed	(iii) Velocity	(iv) Area
Vector	Scalar	Vector	Scalar
(v) Acceleration	(vi) Work	(vii) Volume	(viii) Weight
Vector	Scalar	Scalar	Vector
(ix) Mass	Scalar		

Q.2 Write the vector $\overline{PQ}$ in the form $xi + yj$

- (i) $P(4,6), Q(-4,5)$ (ii) $P\left(-\frac{5}{2}, 9\right), Q(10,8)$
 (iii) $P(-3,6), Q(5,9)$ (ii) $P(-5,2), Q(0,-5)$

SOLUTION:

- (i)
- $P(4,6), Q(-4,5)$

$$\overline{PQ} = Q - P = (-4,5) - (4,6) = (-4-4, 5-6) = (-8, -1)$$

$$\overline{PQ} = -8i - j$$

- (ii)
- $P\left(-\frac{5}{2}, 9\right), Q(10,8)$

$$\overline{PQ} = Q - P = (10,8) - \left(-\frac{5}{2}, 9\right) = \left(10 + \frac{5}{2}, 8 - 9\right)$$

$$= \left(\frac{25}{2}, -1\right) = \frac{25}{2}i - j$$

- (iii)
- $P(-3,6), Q(5,9)$

$$\overline{PQ} = Q - P = (5,9) - (-3,6) = (5+3, 9-6) = (8,3) = 8i + 3j$$

(ii) $P(-6, 2), Q(0, -5)$

$$\overrightarrow{PQ} = Q - P = (0, -5) - (-6, 2) = (0 + 6, -5 - 2) = (6, -7)$$

$$\overrightarrow{PQ} = -6i - 7j$$

Q.3 Find the following vectors if $\vec{p} = -5i + 4j$, $\vec{q} = -3i + 5j$ and $\vec{r} = -6i + 7j$.

(i) $2\vec{p} + 3\vec{q}$

(ii) $3\vec{p} + \frac{1}{2}\vec{q} + \vec{r}$

(iii) $3\vec{p} + 5\vec{q} - 4\vec{r}$

(iv) $\frac{1}{2}\vec{p} - 2\vec{r} + 4\vec{q}$

SOLUTION:

$$\vec{p} = -5i + 4j, \vec{q} = -3i + 5j \text{ and } \vec{r} = -6i + 7j$$

(i) $2\vec{p} + 3\vec{q}$

$$= 2(-5i + 4j) + 3(-3i + 5j)$$

$$= (-10i + 8j) + (-9i + 15j)$$

$$= -19i + 23j$$

(ii) $3\vec{p} + \frac{1}{2}\vec{q} + \vec{r}$

$$= 3(-5i + 4j) + \frac{1}{2}(-3i + 5j) + (-6i + 7j)$$

$$= (-15i + 12j) + \left(-\frac{3}{2}i + \frac{5}{2}j\right) + (-6i + 7j)$$

$$= -\frac{45}{2}i + \frac{43}{2}j$$

(iii) $3\vec{p} + 5\vec{q} - 4\vec{r}$

$$= 3(-5i + 4j) + 5(-3i + 5j) - 4(-6i + 7j)$$

$$= (-15i + 12j) + (-15i + 25j) + (24i - 28j)$$

$$= -6i + 9j$$

$$(iv) \frac{1}{2}\vec{p} - 2\vec{r} + 4\vec{q}$$

$$= \frac{1}{2}(-5i + 4j) - 2(-3i + 5j) + 4(-6i + 7j)$$

$$= \left(-\frac{5}{2}i + 2j\right) + (6i - 10j) + (-24i + 28j)$$

$$= \left(-\frac{5}{2}i + 2j\right) + (-18i + 18j)$$

$$= -\frac{41}{2}i + 20j$$

Q.4 P(-2, 5), Q(8, 0), R($\frac{7}{2}$, 6) and S(-10, 7) are the given points. Find the sum of the vectors $\vec{PQ}$ and $\vec{RS}$.

SOLUTION:

$$\vec{PQ} + \vec{RS} = Q - P + S - R = Q + S - P - R$$

$$= (8, 0) + (-10, 7) - (-2, 5) - \left(\frac{7}{2}, 6\right)$$

$$= \left(8 - 10 + 2 - \frac{7}{2}, 0 + 7 - 5 - 6\right)$$

$$= \left(-\frac{7}{2}, -4\right)$$

$$= -\frac{7}{2}i - 4j$$

Q.5 Find the unit vector in the direction of each of the following vector:

(i) $\vec{p} = -3i + 8j$

(ii) $\vec{q} = 5i - 9j$

(iii) $\vec{r} = 10i - 12j$

(iv) (i) $\vec{s} = 7i + 5j$

SOLUTION:

(i) $\vec{p} = -3i + 8j$

$$\hat{p} = \frac{\vec{p}}{|\vec{p}|} = \frac{-3i + 8j}{\sqrt{(-3)^2 + (8)^2}} = \frac{-3i + 8j}{\sqrt{73}}$$

(ii) $\vec{q} = 5i - 9j$

$$\hat{q} = \frac{\vec{q}}{|\vec{q}|} = \frac{5i - 9j}{\sqrt{(5)^2 + (-9)^2}} = \frac{5i - 9j}{\sqrt{106}}$$

(iii) $\vec{r} = 10i - 12j$

$$\begin{aligned}\hat{r} &= \frac{\vec{r}}{|\vec{r}|} = \frac{10i - 12j}{\sqrt{(10)^2 + (-12)^2}} = \frac{10i - 12j}{\sqrt{244}} = \frac{2(5i - 6j)}{2\sqrt{61}} \\ &= \frac{(5i - 6j)}{\sqrt{61}}\end{aligned}$$

(iv) $\vec{s} = 7i + 5j$

$$\hat{s} = \frac{\vec{s}}{|\vec{s}|} = \frac{7i + 5j}{\sqrt{(7)^2 + (5)^2}} = \frac{7i + 5j}{\sqrt{74}}$$

Q.6 Find the vectors in the direction of the following vectors with given magnitudes:

(i) $\vec{a} = 2i - 3j$, magnitude 6 units

(ii) $\vec{b} = 10i - 9j$, magnitude 7 units

(iii) $\vec{c} = 5i + 6j$, magnitude 11 units

SOLUTION:

(i) $\vec{a} = 2i - 3j$, magnitude 6 units

$$\frac{6\vec{a}}{|\vec{a}|} = \frac{6(2i - 3j)}{\sqrt{(2)^2 + (-3)^2}} = \frac{12i - 18j}{\sqrt{13}}$$

(ii) $\vec{b} = 10i - 9j$, magnitude 7 units

$$\frac{7\vec{b}}{|\vec{b}|} = \frac{7(10i - 9j)}{\sqrt{(10)^2 + (-9)^2}} = \frac{70i - 63j}{\sqrt{181}}$$

(iii) $\vec{c} = 5i + 6j$, magnitude 11 units

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$$\frac{11i}{|c|} = \frac{11(5i+6j)}{\sqrt{(5)^2+(6)^2}} = \frac{55i+66j}{\sqrt{61}}$$

Q.7 Find the values of unknown if:

(i) $4i + 5j = pi + 5j$

(ii) $(5, 7) = (5, -q)$

(iii) $\frac{7}{2}i - \frac{3}{5}j = pi - \frac{3}{\sqrt{25}}j$

(iv) $\begin{pmatrix} \frac{5}{2} \\ -7 \end{pmatrix} = \begin{pmatrix} p \\ -7 \end{pmatrix}$

SOLUTION:

(i) $4i + 5j = pi + 5j$

$p = 4$

(ii) $(5, 7) = (5, -q)$

$-q = 7 \Rightarrow q = -7$

(iii) $\frac{7}{2}i - \frac{3}{5}j = pi - \frac{3}{\sqrt{25}}j$

$p = \frac{7}{2}$

(iv) $\begin{pmatrix} \frac{5}{2} \\ -7 \end{pmatrix} = \begin{pmatrix} p \\ -7 \end{pmatrix}$

$p = \frac{5}{2}$

Q.8 Find the position vector of point dividing the join of A and B in the given ratio, where position vectors of A and B are given:

(i) $4i - 5j$ and $2i + 7j$ where ratio 3:5

(ii) $3i + 5j$ and $11i + 8j$ where ratio 3:2

(iii) $2i - 3j$ and $2i + 6j$ where ratio 4:3

SOLUTION:

(i) $4i - 5j$ and $2i + 7j$ where ratio 3:5

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

$$\vec{r} = \frac{3(2i+7j) + 5(4i-5j)}{3+5} = \frac{(6i+21j) + (20i-25j)}{8} = \frac{26i-4j}{8} = \frac{13i-2j}{4}$$

(ii) $3i + 5j$ and $11i + 8j$ where ratio 3:2

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

$$\vec{r} = \frac{3(11i+8j) + 2(3i+5j)}{3+2} = \frac{(33i+24j) + (6i+10j)}{5} = \frac{39i+34j}{5}$$

(iii) $2i - 3j$ and $2i + 6j$ where ratio 4:3

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

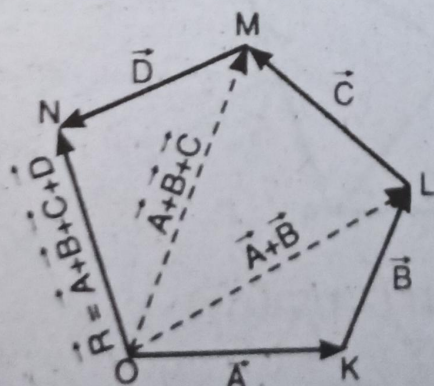
$$\vec{r} = \frac{4(2i+6j) + 3(2i-3j)}{4+3} = \frac{(8i+24j) + (6i-9j)}{7} = \frac{14i+15j}{7}$$

Q.9 Prove Hexagon law of vector addition.

SOLUTION:

"According to law of hexagon, if sides of hexagon represents the magnitude of vectors then closing side of hexagon represent the resultant."

Consider a regular hexagon ABCDEF with O and its center.



In $\triangle OKL$

$$\vec{OL} = \vec{OK} + \vec{KL}$$

$$\overrightarrow{OL} = \vec{A} + \vec{B}$$

In $\triangle OLM$

$$\overrightarrow{OM} = \overrightarrow{OL} + \overrightarrow{LM}$$

$$\overrightarrow{OM} = \vec{A} + \vec{B} + \vec{C}$$

In $\triangle OMN$

$$\overrightarrow{ON} = \overrightarrow{OM} + \overrightarrow{MN}$$

$$\overrightarrow{ON} = \vec{A} + \vec{B} + \vec{C} + \vec{D}$$

Q.10 If $A(3, -5)$, $B(6, 0)$ and $C(2, 4)$ be vertices of a parallelogram, then using vectors. Find the coordinates of point D if:

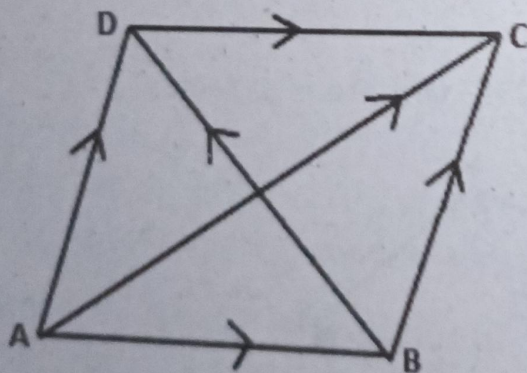
(i) $ABCD$ is a parallelogram

(ii) $ADBC$ is a parallelogram

(iii) $ABDC$ is a parallelogram

SOLUTION:

(i) $ABCD$ is a parallelogram



$A(3, -5)$, $B(6, 0)$ and $C(2, 4)$, $D(x, y)$

$$\overrightarrow{OD} = (x, y) - (0, 0) = (x, y)$$

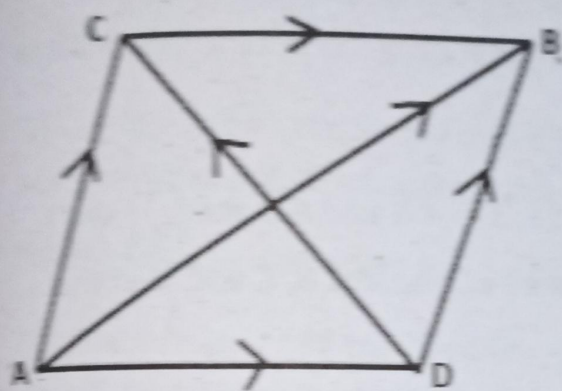
$$\overrightarrow{OA} = (3, -5) - (0, 0) = (3, -5)$$

$$\overrightarrow{AD} = \overrightarrow{BC} = \vec{C} - \vec{B} = (2, 4) - (6, 0) = (2 - 6, 4) = (-4, 4)$$

$$\overrightarrow{OD} = \overrightarrow{OA} + \overrightarrow{AD}$$

$$\overrightarrow{OD} = (3, -5) + (-4, 4) = (3 - 4, -5 + 4) = (-1, -1)$$

(ii) ADBC is a parallelogram



$A(3, -5), B(6, 0)$ and $C(2, 4), D(x, y)$

$$\overrightarrow{DB} = \overrightarrow{AC}$$

$$\vec{B} - \vec{D} = \vec{C} - \vec{A}$$

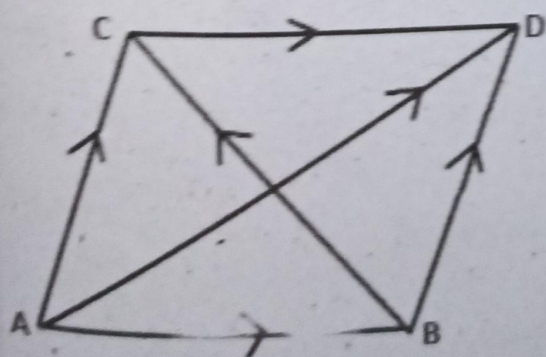
$$(6, 0) - (x, y) = (2, 4) - (3, -5)$$

$$(6 - x, -y) = (-1, 9)$$

$$6 - x = -1, -y = 9$$

$$x = 7, y = -9$$

(iii) ABDC is a parallelogram



$A(3, -5), B(6, 0)$ and $C(2, 4), D(x, y)$

$$\overrightarrow{BD} = \overrightarrow{AC}$$

$$\vec{D} - \vec{B} = \vec{C} - \vec{A}$$

$$(x, y) - (6, 0) = (2, 4) - (3, -5)$$

$$(x - 6, y) = (-1, 9)$$

$$x - 6 = -1, y = 9$$

$$x = 5, y = -9$$

Q.11 Given that A is the point (1, 3). $\vec{AB}$ and $\vec{AD}$ are $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ respectively. Find the coordinates of vertices B, C and D of the parallelogram ABCD.

SOLUTION:

$$A(1, 3)$$

$$\vec{AB} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \text{ and } \vec{AD} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\vec{AB} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\vec{B} - \vec{A} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\vec{B} = \vec{A} + \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\vec{B} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\vec{B} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

$$\vec{AD} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\vec{D} - \vec{A} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\vec{D} = \vec{A} + \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\vec{D} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

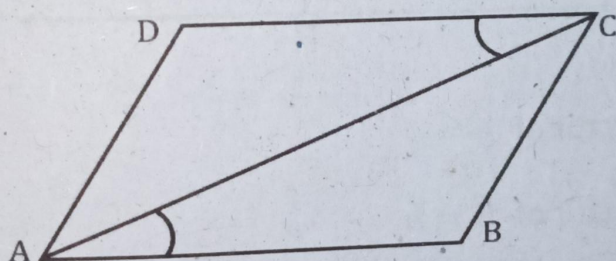
$$\vec{D} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

Q.12 Using vectors, prove that:

- (i) If one pair of opposite sides of a quadrilateral are congruent and parallel then it is a parallelogram.
- (ii) The line segment joining the mid-points of the opposite sides of a quadrilateral bisect each other.
- (iii) Medians of a triangle are concurrent.
- (iv) The mid-point of the hypotenuse of a right-angled triangle is equidistant from its vertices.

SOLUTION:

(i)



$$\text{Given } \overline{AB} = \overline{CD}$$

$$\text{Given } \overline{AD} = \overline{BC}$$

$$\triangle ABC \leftrightarrow \triangle CDA$$

$$\overline{AB} = \overline{CD}$$

$$\overline{AD} = \overline{BC}$$

$$\overline{AC} = \overline{AC}$$

$$\triangle ABC \cong \triangle CDA$$

$$\angle ACD \cong \angle BAC$$

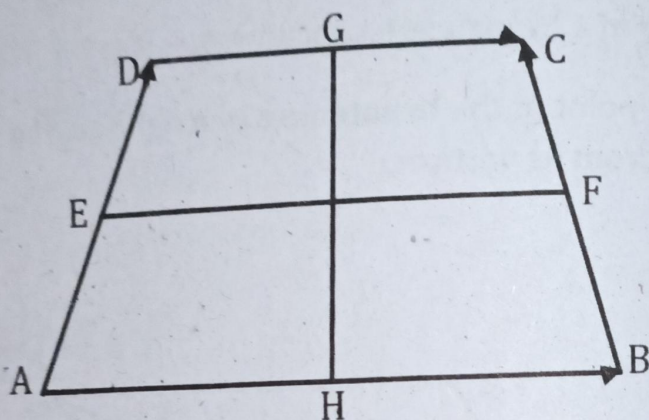
$$\angle DAC \cong \angle ACB$$

The sides are parallel as alternate angles are equal.

$$AB \parallel CD \text{ and } AD \parallel BC$$

The pairs of opposite sides of ABCD are parallel therefore ABCD is a parallelogram.

(ii)



Let position vector of $\vec{A} = \vec{a}$

Let position vector of $\vec{B} = \vec{b}$

Let position vector of $\vec{C} = \vec{c}$

Let position vector of $\vec{D} = \vec{d}$

Position vector of $\vec{AB} = \vec{a}$

Position vector of $\vec{BC} = \vec{b} - \vec{a}$

Position vector of $\vec{DC} = \vec{b} - \vec{c}$

Position vector of $\vec{AD} = \vec{c}$

Position vector of $\vec{F} = \frac{\vec{a} + \vec{b}}{2}$

position vector of $\vec{F} = \frac{\vec{c}}{2}$

position vector of $\vec{H} = \frac{\vec{a}}{2}$

position vector of $\vec{G} = \frac{\vec{b} + \vec{c}}{2}$

position vector of $\overrightarrow{GH} = \vec{a} - \frac{\vec{b} + \vec{c}}{2}$

$$= \frac{2\vec{a} - \vec{b} - \vec{c}}{2}$$

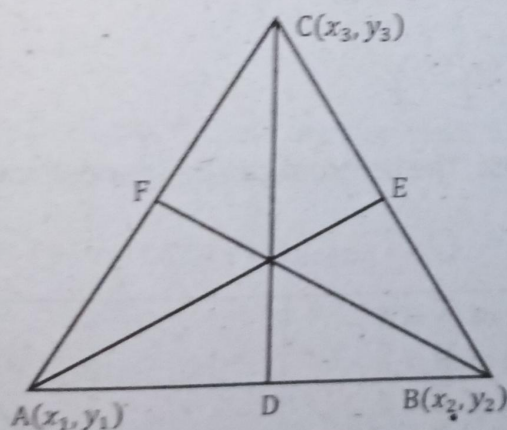
Position vector of $\overrightarrow{EF} = \frac{\vec{a} + \vec{b}}{2} - \frac{\vec{c}}{2}$

$$= \frac{\vec{a} + \vec{b} - \vec{c}}{2}$$

Mid-point of $\overrightarrow{GH} = \frac{\vec{a}}{2} + \frac{\vec{b} + \vec{c}}{2} = \frac{\vec{a} + \vec{b} + \vec{c}}{2}$

Mid-point of $\overrightarrow{EF} = \frac{\vec{a} + \vec{b}}{2} + \frac{\vec{c}}{2} = \frac{\vec{a} + \vec{b} + \vec{c}}{2}$

(iii)



Let

G be the centroid

G divides $\overline{AE}$ in 2:1

G divides $\overline{BF}$ in 2: 1

G divides $\overline{CD}$ in 2: 1

$$A = (x_1, y_1)$$

$$B = (x_2, y_2)$$

$$C = (x_3, y_3)$$

$$D = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$$

$$E = \left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2} \right)$$

$$F = \left(\frac{x_1+x_3}{2}, \frac{y_1+y_3}{2} \right)$$

$$\left(\frac{k_1x_2+k_2x_1}{k_1+k_2}, \frac{k_1y_2+k_2y_1}{k_1+k_2} \right)$$

Applying ratiom theorem in $\overline{AE}$

$$G = \left(\frac{1(x_1)+2\left(\frac{x_2+x_3}{2}\right)}{1+2}, \frac{1(y_1)+2\left(\frac{y_2+y_3}{2}\right)}{1+2} \right)$$

$$G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$$

Using same process, the centroid can be found. Proved

(iv)

Let Position vector of $\vec{A} = \vec{a}$

Let Position vector of $\vec{B} = \vec{b}$

Let Position vector of $\vec{C} = \vec{c}$

Let Position vector of $\vec{D} = \vec{d}$

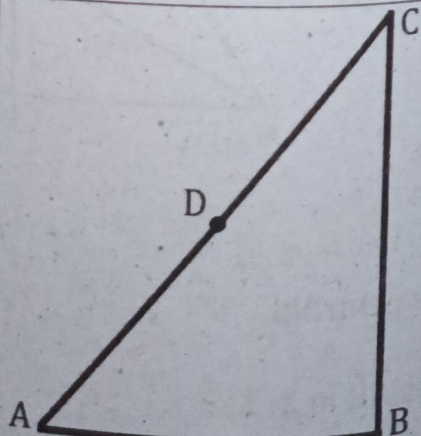
Position vector of $\overline{AD} =$

Position vector of $\overline{DC}$

$$\vec{d} - \vec{a} = \vec{c} - \vec{d}$$

$$\vec{d} + \vec{d} = \vec{a} + \vec{c}$$

$$2\vec{d} = \vec{a} + \vec{c}$$



$$\vec{a} = \frac{\vec{a} + \vec{c}}{2}$$

Position vector of $\overrightarrow{AD} = \vec{D} - \vec{A}$

$$= \frac{\vec{a} + \vec{c}}{2} - \vec{a}$$

$$= \frac{\vec{a} + \vec{c} - 2\vec{a}}{2}$$

$$= \frac{\vec{c} - \vec{a}}{2}$$

Position vector of $\overrightarrow{DC} = \vec{C} - \vec{D}$

$$= \vec{c} - \frac{\vec{a} + \vec{c}}{2}$$

$$= \frac{2\vec{c} - \vec{a} - \vec{c}}{2}$$

$$= \frac{\vec{c} - \vec{a}}{2}$$

Q.13 Given that the points $(1, 1)$, $(5, 4)$, $(8, 9)$ and $(0, 3)$ represent position vectors of A, B, C and D respectively. Show that ABCD is a trapezium.

SOLUTION:

We know that in a trapezium, there are two sides which are parallel while other two are non-parallel.

i.e. We have to prove that two of the given vectors are parallel only.

A(1,1), B(5,4), C(8,9) and D(0,3)

$$\overrightarrow{AB} = (5,4) - (1,1) = (4,3)$$

$$\overrightarrow{BC} = (8,9) - (5,4) = (3,5)$$

$$\overrightarrow{CD} = (0,3) - (8,9) = (-8,-6)$$

Take -2 common

$$\overrightarrow{CD} = -2(4,3)$$

$$\overrightarrow{CD} = -2\overrightarrow{AB}$$

$$\overrightarrow{DA} = (1,1) - (0,3) = (1, -2)$$

Vector $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel. Proved

Q.14 A, C are the points $\vec{a}, \vec{b}$ and $2\vec{a} - \vec{b}$ respectively. D divides $\overrightarrow{AC}$ in 2: 3 and E divides $\overrightarrow{BD}$ in 4: 1. Find the position vector of E.

SOLUTION:

$\vec{a}, \vec{b}$ and $2\vec{a} - \vec{b}$ are position vectors of A, B and C respectively.

$\therefore$ D divides $\overrightarrow{AC}$ in ratio 2: 3 then by ratio theorem $r = \frac{lA+mB}{l+m}$

$$D = \frac{3a+2(2a-b)}{2+3}$$

$$D = \frac{3a+4a-2b}{5}$$

$$D = \frac{7a-2b}{5}$$

$\therefore$ E divides $\overrightarrow{BD}$ in ratio 4: 1 then by ratio theorem $r = \frac{lA+mB}{l+m}$

$$E = \frac{mB+lD}{l+m}$$

$$E = \frac{1}{1+4} \left\{ 1(b) + 4 \left(\frac{7a-2b}{5} \right) \right\}$$

$$E = \frac{1}{5} \left\{ \frac{5b+28a-8b}{5} \right\}$$

$$E = \frac{28a-3b}{5(5)}$$

$$E = \frac{28a-3b}{25}$$

Q.15 In a parallelogram ABCD, X is the mid-point of $\overline{AB}$ and Y divides $\overline{BC}$ in 1: 2. Show that if Z divides $\overline{DX}$ in 6: 1 then also divide $\overline{AY}$ in 3: 4.

SOLUTION:

a, b, c and d position vector of A, B, C and D respectively in parallelogram ABCD

$$b - a = c - d$$

$$\text{or } \boxed{c = d + b - a}$$

The mid-point of $\overline{AB}$ is X then by ratio theorem

$$X = \frac{a + b}{2}$$

$\therefore$ Y divides $\overline{BC}$ in ratio 1: 2 then by ratio theorem $r = \frac{lA+mB}{l+m}$

$$y = \frac{2b+1 \times c}{1+2}$$

$$y = \frac{2b+c}{3}$$

$\therefore$ Z divides $\overline{DX}$ in ratio 6: 1 then by ratio theorem $r = \frac{lA+mB}{l+m}$

$$Z = \frac{1 \times D + 6X}{6+1}$$

$$Z = \frac{1}{7} \left\{ 1(d) + 6 \left(\frac{a+b}{2} \right) \right\}$$

$$Z = \frac{d+3(a+b)}{7}$$

$$Z = \frac{d+3a+3b}{7} \rightarrow (A)$$

$\therefore$ Z divides $\overline{AY}$ in ratio 3: 4 then by ratio theorem $r = \frac{lA+mB}{l+m}$

$$Z = \frac{4(A)+3(y)}{7}$$

$$Z = \frac{1}{7} \left\{ 4(a) + 3 \left(\frac{2b+c}{3} \right) \right\}$$

$$Z = \frac{4a+2b+c}{7}$$

$$\text{As } c = d + b - a$$

$$Z = \frac{4a+2b+d+b-a}{7}$$

$$Z = \frac{a+3b+d}{7} \rightarrow (B)$$

From Eq (A) and Eq (B)

Hence it is proved that Z divides $\overrightarrow{DX}$ in 6:1 and divides $\overrightarrow{AY}$ in 3:4

EXERCISE 3.2

Q.1 Let $\vec{u} = i + 2j + k, \vec{v} = 3i - 5j, \vec{w} = -8i + 7j - 2k$. Find:

(i) $3\vec{u} + 2\vec{v}$ (ii) $\vec{v} - 2\vec{u}$ (iii) $\vec{u} - 3\vec{v} + 2\vec{w}$

(iv) $3\vec{u} + \vec{v} - \vec{w}$ (v) $-2\vec{u} + \frac{1}{2}\vec{v} - 3\vec{w}$

SOLUTION:

$$\vec{u} = i + 2j + k, \vec{v} = 3i - 5j, \vec{w} = -8i + 7j - 2k$$

(i) $3\vec{u} + 2\vec{v}$

$$= 3(i + 2j + k) + 2(3i - 5j)$$

$$= 9i - 4j + 3k$$

(ii) $\vec{v} - 2\vec{u}$

$$= 3i - 5j - 2(i + 2j + k)$$

$$= i - 9j - 2k$$

(iii) $\vec{u} - 3\vec{v} + 2\vec{w}$

$$= (i + 2j + k) - 3(3i - 5j) + 2(-8i + 7j - 2k)$$

$$= -24i + 31j - 3k$$

(iv) $3\vec{u} + \vec{v} - \vec{w}$

$$= 3(i + 2j + k) + (3i - 5j) - (-8i + 7j - 2k)$$

$$= (3i + 6j + 3k) + (3i - 5j) + 8i - 7j + 2k$$

$$= 14i - 6j + 5k$$

(v) $-2\vec{u} + \frac{1}{2}\vec{v} - 3\vec{w}$

$$= -2(i + 2j + k) + \frac{1}{2}(3i - 5j) - 3(-8i + 7j - 2k)$$

$$= (-2i - 4j - 2k) + \left(\frac{3}{2}i - \frac{5}{2}j\right) + (24i - 21j + 6k)$$

$$= (22i - 25j + 4k) + \left(\frac{3}{2}i - \frac{5}{2}j\right)$$

$$= \frac{47}{2}i - \frac{55}{2}j + 4k$$

Q.2 If $\vec{a} = -5i + 3j - 4k$ then find:

(i) $-2\vec{a}$

(ii) $3|\vec{a}|$

(iii) $\frac{4}{5}\vec{a}$

(iv) $-\frac{1}{2}\vec{a}$

SOLUTION:

(i) $-2\vec{a} = -2(-5i + 3j - 4k) = 10i - 6j + 8k$

(ii) $3|\vec{a}| = 3\sqrt{(-5)^2 + (3)^2 + (-4)^2} = 3\sqrt{50} = 3\sqrt{25(2)}$

$$3|\vec{a}| = 3(5)\sqrt{2} \Rightarrow 3|\vec{a}| = 15\sqrt{2}$$

(iii) $\frac{4}{5}\vec{a} = \frac{4}{5}(-5i + 3j - 4k) = -4i + \frac{12}{5}j - \frac{16}{5}k$

(iv) $-\frac{1}{2}\vec{a} = -\frac{1}{2}(-5i + 3j - 4k) = \frac{5i}{2} - \frac{3j}{2} + \frac{4k}{2} = \frac{5i}{2} - \frac{3j}{2} + k$

Q.3 Let $\vec{A} = i + 2j - 3k$, $\vec{B} = 2i - j + 2k$, $\vec{C} = 3i - j + 5k$. Find

(i) $\vec{A} - 2\vec{B}$

(ii) $3\vec{B} + 2\vec{C}$

(iii) $3\vec{A} - (2\vec{B} + \vec{C})$

SOLUTION:

(i) $\vec{A} - 2\vec{B} = (i + 2j - 3k) - 2(2i - j + 2k)$

$$= (i + 2j - 3k) + (-4i + 2j - 4k)$$

$$= -3i + 4j - 7k$$

(ii) $3\vec{B} + 2\vec{C} = 3(2i - j + 2k) + 2(3i - j + 5k)$

$$= (6i - 3j + 6k) + (6i - 2j + 10k)$$

$$= 12i - 5j + 16k$$

$$(iii) 3\vec{A} - (2\vec{B} + \vec{C}) = 3\vec{A} - 2\vec{B} - \vec{C}$$

$$= 3(i + 2j - 3k) - 2(2i - j + 2k) - (3i - j + 5k)$$

$$= (3i + 6j - 9k) + (-4i + 2j - 4k) - 3i + j - 5k$$

$$= -4i + 9j - 18k$$

Q.4 Let $\vec{u} = i - 3j + 2k$, $\vec{v} = i + j$ and $\vec{w} = 2i + 2j - 4k$. Find

$$(i) |\vec{u} + \vec{v} - \vec{w}|$$

$$(ii) |\vec{u}| + |\vec{v}|$$

$$(iii) \left| \frac{1}{|\vec{w}|} \vec{w} \right|$$

SOLUTION:

$$(i) |\vec{u} + \vec{v} - \vec{w}|$$

$$= |(i - 3j + 2k) + (i + j) - (2i + 2j - 4k)|$$

$$= |(2i - 2j + 2k) - 2i - 2j + 4k|$$

$$= |-4j + 6k|$$

$$= \sqrt{(-4)^2 + (6)^2}$$

$$= \sqrt{52}$$

$$(ii) |\vec{u}| + |\vec{v}|$$

$$= \sqrt{(1)^2 + (-3)^2 + (2)^2} + \sqrt{(1)^2 + (1)^2} = \sqrt{14} + \sqrt{2}$$

$$(iii) \left| \frac{1}{|\vec{w}|} \vec{w} \right|$$

$$|\vec{w}| = \sqrt{(2)^2 + (2)^2 + (-4)^2} = \sqrt{24} = \sqrt{6(4)} = 2\sqrt{6}$$

$$\left| \frac{1}{|\vec{w}|} \vec{w} \right| = \left| \frac{1}{2\sqrt{6}} (2i + 2j - 4k) \right| = \frac{1}{2\sqrt{6}} |(2i + 2j - 4k)|$$

$$= \frac{1}{2\sqrt{6}} \sqrt{(2)^2 + (2)^2 + (-4)^2} = \frac{1}{2\sqrt{6}} (2\sqrt{6}) = 1$$

Q.5 (a) The position vector of point P is $\begin{pmatrix} 5 \\ -3 \\ 2 \end{pmatrix}$ and the position vector of point Q is $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$. Find $4\overrightarrow{PQ}$ and $|5\overrightarrow{PQ}|$.

(b) If $\vec{a} = 3i - 5j + 2k$, $\vec{b} = -i - 2j + 3k$ then find

- (i) $\hat{a}$ (ii) $\hat{b}$ (iii) $|\vec{a} + \vec{b}|$ (iv) $|\vec{a} - \vec{b}|$

SOLUTION:

$$(i) \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{3i - 5j + 2k}{\sqrt{(3)^2 + (-5)^2 + (2)^2}} = \frac{3i - 5j + 2k}{\sqrt{38}}$$

$$(ii) \hat{b} = \frac{\vec{b}}{|\vec{b}|} = \frac{-i - 2j + 3k}{\sqrt{(-1)^2 + (-2)^2 + (3)^2}} = \frac{-i - 2j + 3k}{\sqrt{14}}$$

$$(iii) |\vec{a} + \vec{b}| = |3i - 5j + 2k - i - 2j + 3k| = |2i - 7j + 5k|$$

$$= \sqrt{(2)^2 + (-7)^2 + (5)^2} = \sqrt{78}$$

$$(iv) |\vec{a} - \vec{b}| = |3i - 5j + 2k - (-i - 2j + 3k)| = |4i - 3j - k|$$

$$= \sqrt{(4)^2 + (-3)^2 + (-1)^2} = \sqrt{26}$$

Q.6 Find the vector A_1A_2 when:

(i) $A_1(0,0,0), A_2(-2,5,1)$ (ii) $A_1(2,1,-3), A_2(7,1,-3)$

(iii) $A_1(5,-2,1), A_2(2,4,2)$

SOLUTION:

(i) $A_1(0,0,0), A_2(-2,5,1)$

$$A_1A_2 = A_2 - A_1 = (-2,5,1) - (0,0,0) = (-2,5,1)$$

(ii) $A_1(2,1,-3), A_2(7,1,-3)$

$$A_1A_2 = A_2 - A_1 = (7,1,-3) - (2,1,-3) = (5,0,0)$$

(iii) $A_1(5,-2,1), A_2(2,4,2)$

$$A_1 A_2 = A_2 - A_1 = (2, 4, 2) - (5, -2, 1) = (-3, 6, 1)$$

Q.7 (i) Find the initial point of the vector $\vec{a} = (-2, 1, 2)$ if the terminal point is $(4, 0, -1)$.

(ii) Find the terminal point of the vector $\overrightarrow{A_1 A_2} = (1, 3, -3)$ if the initial point is $(-2, 1, 4)$.

SOLUTION:

(i) Find the initial point of the vector $\vec{a} = (-2, 1, 2)$ if the terminal point is $(4, 0, -1)$.

Let $\vec{r}$ be the initial point.

$$(-2, 1, 2) = (4, 0, -1) - \vec{r}$$

$$\vec{r} = (4, 0, -1) - (-2, 1, 2) = (4 + 2, 0 - 1, -1 - 2) = (6, -1, -3)$$

(ii) Find the terminal point of the vector $\overrightarrow{A_1 A_2} = (1, 3, -3)$ if the initial point is $(-2, 1, 4)$.

$$\overrightarrow{A_1 A_2} = (1, 3, -3)$$

$$A_2 - A_1 = (1, 3, -3) \Rightarrow A_2 = A_1 + (1, 3, -3)$$

$$A_2 = (-2, 1, 4) + (1, 3, -3) \Rightarrow A_2 = (-2 + 1, 1 + 3, 4 - 3)$$

$$A_2 = (-1, 4, 1)$$

Q.8 The initial point of a vector $\vec{a}$ of magnitude 5 is $(1, -\sqrt{3}, -5)$. Find k if the terminal point is:

(a) $(3, \sqrt{3}, k)$

(b) $(2, -3\sqrt{3}, -10k)$

(c) $(-2, k, -3)$

SOLUTION:

(a) $(3, \sqrt{3}, k)$

$$\vec{a} = (3, \sqrt{3}, k) - (1, -\sqrt{3}, -5)$$

$$\vec{a} = (3 - 1, \sqrt{3} + \sqrt{3}, k + 5) = (2, 2\sqrt{3}, k + 5)$$

$$|\vec{a}| = \sqrt{(2)^2 + (2\sqrt{3})^2 + (k+5)^2}$$

$$5^2 = 4 + 12 + (k+5)^2 \Rightarrow (k+5)^2 = 25 - 16 \Rightarrow (k+5)^2 = 9$$

$$k+5 = \pm 3 \Rightarrow k = -5 + 3, -5 - 3 \Rightarrow \boxed{k = -2, -8}$$

$$(b) (2, -3\sqrt{3}, -10k)$$

$$\vec{a} = (2, -3\sqrt{3}, -10k) - (1, -\sqrt{3}, -5)$$

$$\vec{a} = (2-1, -3\sqrt{3} + \sqrt{3}, -10k+5) = (1, -2\sqrt{3}, 5-10k)$$

$$|\vec{a}| = \sqrt{(1)^2 + (-2\sqrt{3})^2 + (5-10k)^2}$$

$$5^2 = 1 + 12 + (5-10k)^2 \Rightarrow (5-10k)^2 = 25 - 13$$

$$(5-10k)^2 = 12 \Rightarrow 5-10k = \pm\sqrt{12} \Rightarrow 5 \pm 2\sqrt{3} = 10k$$

$$\boxed{k = \frac{5 \pm 2\sqrt{3}}{10}}$$

$$(c) (-2, k, -3)$$

$$\vec{a} = (-2, k, -3) - (1, -\sqrt{3}, -5) = (-2-1, k+\sqrt{3}, -3+5)$$

$$\vec{a} = (-3, k+\sqrt{3}, 2)$$

$$5^2 = (-3)^2 + (k+\sqrt{3})^2 + (2)^2 \Rightarrow 25 - 9 - 4 = (k+\sqrt{3})^2$$

$$(k+\sqrt{3})^2 = 12 \Rightarrow k+\sqrt{3} = \pm\sqrt{12} \Rightarrow k+\sqrt{3} = \pm 2\sqrt{3}$$

$$k = -\sqrt{3} \pm 2\sqrt{3} \Rightarrow k = -\sqrt{3} + 2\sqrt{3}, -\sqrt{3} - 2\sqrt{3}$$

$$\boxed{k = \sqrt{3}, -3\sqrt{3}}$$

Q.9 Find the coordinates of a, if $\vec{OA}$ is of length 6 units in the direction of $\vec{OB}$, where B is the point $(2, -1, 4)$.

SOLUTION:

$$\overrightarrow{OA} = \frac{6\overrightarrow{OB}}{|\overrightarrow{OB}|} = \frac{6(2i-j+4k)}{\sqrt{(2)^2+(-1)^2+(4)^2}} = \frac{12i-6j+24k}{\sqrt{21}}$$

Q.10 P, Q, R, S are the points with position vectors given by $(1,1,-1)$, $(1,-1,2)$, $(0,1,1)$ and $(2,1,0)$ respectively.

(i) Find $|\overrightarrow{PQ}|$ and $|\overrightarrow{QS}|$

(ii) Find the position vector of the point which:

(a) divides $\overline{QR}$ internally in the ratio 3: 2

(b) divides $\overline{PR}$ externally in the ratio 3: 2

(iii) If X and Y are the midpoints of $\overline{PR}$ and $\overline{RS}$. Show that $\overline{XY} = \frac{1}{2}\overline{PS}$.

SOLUTION:

P(1,1,-1), Q(1,-1,2), R(0,1,1) and S(2,1,0)

(i) $|\overrightarrow{PQ}|$

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= (1,-1,2) - (1,1,-1)$$

$$= (0,-2,3)$$

$$|\overrightarrow{PQ}| = \sqrt{(0)^2 + (-2)^2 + (3)^2} = \sqrt{13}$$

$|\overrightarrow{QS}|$

$$\overrightarrow{QS} = \overrightarrow{OS} - \overrightarrow{OQ}$$

$$= (2,1,0) - (1,-1,2)$$

$$= (1,2,-2)$$

$$|\overrightarrow{QS}| = \sqrt{(1)^2 + (2)^2 + (-2)^2} = \sqrt{9} = 3$$

$$(ii) (a) \vec{r} = \frac{3\vec{R}+2\vec{Q}}{3+2} = \frac{3(j+k)+2(i-j+2k)}{5} = \frac{(3j+3k)+(2i-2j+4k)}{5}$$

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 $\vec{r} = \frac{2i+j+7k}{5}$

$$(ii) \vec{r} = \frac{3R-2P}{3-2} = \frac{3(j+k)-2(i+j-k)}{1} = \frac{(3j+3k)+(-2i-2j+2k)}{1}$$

$$= (-2i + j + 5k)$$

If X and Y are the midpoints of $\overline{PR}$ and $\overline{RS}$. Show that $\overrightarrow{XY} = \frac{1}{2} \overrightarrow{PS}$.

$$(iii) X = \frac{1}{2} \{ (1, 1, -1) + (0, 1, 1) \} =$$

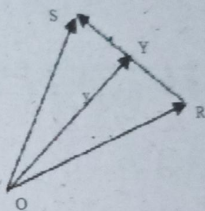
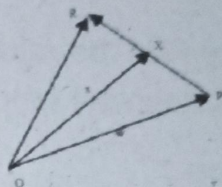
$$\frac{1}{2} (1+0, 1+1, -1+1)$$

$$X = \frac{1}{2} (1, 2, 0) = \left(\frac{1}{2}, 1, 0 \right)$$

$$Y = \frac{1}{2} \{ (0, 1, 1) + (2, 1, 0) \}$$

$$= \frac{1}{2} (0+2, 1+1, 1+0)$$

$$Y = \frac{1}{2} (2, 2, 1) = \left(1, 1, \frac{1}{2} \right)$$



$$\overrightarrow{PS} = (2, 1, 0) - (1, 1, -1) = (2-1, 1-1, 0+1) = (1, 0, 1)$$

$$\overrightarrow{XY} = \vec{Y} - \vec{X} = \left(1, 1, \frac{1}{2} \right) - \left(\frac{1}{2}, 1, 0 \right) = \left(1 - \frac{1}{2}, 1 - 1, \frac{1}{2} - 0 \right)$$

$$\overrightarrow{XY} = \left(\frac{1}{2}, 0, \frac{1}{2} \right) = \frac{1}{2} (1, 0, 1) \Rightarrow \boxed{\overrightarrow{XY} = \frac{1}{2} \overrightarrow{PS}}$$

Q.11 Find the vector $\overrightarrow{OA}$; where:

(i) $|\overrightarrow{OA}| = 6$ and $\overrightarrow{OA}$ is in the direction of the vector $(2, -3, 6)$.

(ii) $|\overrightarrow{OA}| = 2$ and $\overrightarrow{OA}$ is in the opposite direction of the vector $(8, 1, -4)$.

SOLUTION:

$$(i) \overrightarrow{OA} = \frac{6(2i-3j+6k)}{|(2i-3j+6k)|} = \frac{6(2i-3j+6k)}{\sqrt{(2)^2+(-3)^2+(6)^2}} = \frac{12i-18j+36k}{\sqrt{49}}$$

$$= \frac{12i - 18j + 36k}{7}$$

$$(ii) \vec{OA} = \frac{2(8, 1, -4)}{|(8i + j - 4k)|} = \frac{(16i + 2j - 8k)}{\sqrt{(8)^2 + (1)^2 + (-4)^2}} = \frac{16i + 2j - 8k}{\sqrt{81}} = \frac{16i + 2j - 8k}{9}$$

Q.12 (a) Find the magnitude of a vector $\vec{a} = -2i + 3j - 4k$.

SOLUTION:

$$|\vec{a}| = |-2i + 3j - 4k| = \sqrt{(-2)^2 + (3)^2 + (-4)^2} = \sqrt{29}$$

Q.12 (b) Find the value of t if the vectors $-3i + 5j - 6k$ and $2i - 3j + tk$ have the same magnitude.

SOLUTION:

$$|-3i + 5j - 6k| = |2i - 3j + tk|$$

$$\sqrt{(-3)^2 + (5)^2 + (-6)^2} = \sqrt{(2)^2 + (-3)^2 + (t)^2}$$

$$70 = 13 + t^2$$

$$t^2 = 70 - 13 = 57 \Rightarrow t = \pm\sqrt{57}$$

Q.12 (c) Find the vector whose magnitude is 5 times the length of $\vec{a}$ and is in the opposite direction of $\vec{a} = 4i - 5j + 6k$.

SOLUTION:

$$|\vec{a}| = |4i - 5j + 6k| = \sqrt{(4)^2 + (-5)^2 + (6)^2} = \sqrt{77}$$

$$\vec{r} = \frac{-5\vec{a}}{|\vec{a}|} = \frac{-5(4i - 5j + 6k)}{\sqrt{77}} = \frac{-20i + 25j - 30k}{\sqrt{77}}$$

Q.12 (d) Express $\vec{a} = -2i + 4j - 6k$ as a product of its magnitude and direction.

SOLUTION:

$$\vec{a} = -2i + 4j - 6k$$

$$|\vec{a}| = |-2i + 4j - 6k| = \sqrt{(-2)^2 + (4)^2 + (-6)^2} = 2\sqrt{14}$$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{-2i+4j-6k}{2\sqrt{14}} = \frac{-2i}{2\sqrt{14}} + \frac{4j}{2\sqrt{14}} - \frac{6k}{2\sqrt{14}} = \frac{-i}{\sqrt{14}} + \frac{2j}{\sqrt{14}} - \frac{3k}{\sqrt{14}}$$

$$\vec{a} = |\vec{a}|\hat{a} = 2\sqrt{14} \left(\frac{-i}{\sqrt{14}} + \frac{2j}{\sqrt{14}} - \frac{3k}{\sqrt{14}} \right)$$

Q.13 Find the length of the median through point O of the triangle OCD, if C and D are (2, 7, -1) and (4, 1, 2) respectively.

SOLUTION:

$$\text{Mid-point of } \overline{CD} = \frac{1}{2} \{ (2, 7, -1) + (4, 1, 2) \}$$

$$= \frac{1}{2} (2+4, 7+1, -1+2) = \frac{1}{2} (6, 8, 1)$$

$$\text{Length} = \frac{1}{2} |(6, 8, 1)| = \frac{1}{2} \sqrt{(6)^2 + (8)^2 + (1)^2} = \frac{1}{2} \sqrt{101}$$

Q.14 If S and T are the mid points of $\overrightarrow{PR}$ and $\overrightarrow{QR}$ then show that $\overrightarrow{ST} = \frac{1}{2} \overrightarrow{PQ}$.

SOLUTION:

$$\overrightarrow{OP} = a, \overrightarrow{OR} = b, \overrightarrow{OQ} = c$$

$$\vec{S} = \frac{1}{2} (\overrightarrow{OP} + \overrightarrow{OR}) = \frac{1}{2} (a + b)$$

$$\vec{T} = \frac{1}{2} (\overrightarrow{OQ} + \overrightarrow{OR}) = \frac{1}{2} (c + b)$$

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = c - a$$

$$\overrightarrow{ST} = \overrightarrow{OT} - \overrightarrow{OS} = \frac{1}{2} (c + b) - \frac{1}{2} (a + b) = \frac{1}{2} (c + b - a - b)$$

$$\overrightarrow{ST} = \frac{1}{2} (c - a) \Rightarrow \boxed{\overrightarrow{ST} = \frac{1}{2} \overrightarrow{PQ}}$$

EXERCISE 3.3

Q.1 If $\vec{a} = -5i + 3j - 4k$ and $\vec{b} = 2i + 4j + 5k$, then:

(i) verify $\vec{0} + \vec{a} = \vec{a} + \vec{0} = \vec{a}$

(ii) Find additive inverse of $\vec{a}$, $\vec{b}$ and $2\vec{a} + 3\vec{b}$

(iii) verify commutative law for vector addition

SOLUTION:

$$(i) 0 - 5i + 3j - 4k = -5i + 3j - 4k + 0 = -5i + 3j - 4k$$

$$(ii) \text{ Additive inverse of } \vec{a} = -(-5i + 3j - 4k) = 5i - 3j + 4k$$

$$\text{Additive inverse of } \vec{b} = -(2i + 4j + 5k) = -2i - 4j - 5k$$

$$\text{Additive inverse of } 2\vec{a} + 3\vec{b} = -(2\vec{a} + 3\vec{b})$$

$$= -2\vec{a} - 3\vec{b}$$

$$= -2(-5i + 3j - 4k) - 3(2i + 4j + 5k)$$

$$= (10i - 6j + 8k) - 6i - 12j - 15k$$

$$= 4i - 18j - 7k$$

(iii) verify commutative law for vector addition

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

$$(-5i + 3j - 4k) + (2i + 4j + 5k)$$

$$= (2i + 4j + 5k) + (-5i + 3j - 4k)$$

$$-3i + 7j + k = -3i + 7j + k$$

Q.2 Let $\vec{u} = 3i + j + 5k$, $\vec{v} = i - 2j$ and $\vec{w} = -i + 3j - k$. Verify associative law of vector addition.

SOLUTION:

$$\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$$

$$3i + j + 5k + (i - 2j - i + 3j - k)$$

$$= (3i + j + 5k + i - 2j) + -i + 3j - k$$

$$3i + 2j + 4k = 3i + 2j + 4k$$

Q.3 Let $\vec{a} = 3i + 2j + 5k$ is a vector and $m = 5$ is a scalar then verify commutative law of scalar multiplication.

SOLUTION:

$$m(\vec{a}) = (\vec{a})m$$

$$5(3i + 2j + 5k) = (3i + 2j + 5k)(5)$$

$$15i + 10j + 25k = 15i + 10j + 25k$$

Q.4 Let $\vec{a} = 5i + 2j - 3k$ is a vector and $m = 2$ and $n = 5$ then verify associative law of scalar multiplication.

SOLUTION:

$$(m + n)(\vec{a}) = m(\vec{a}) + n(\vec{a})$$

$$(2 + 5)(5i + 2j - 3k) = 2(5i + 2j - 3k) + 5(5i + 2j - 3k)$$

$$7(5i + 2j - 3k) = (10i + 4j - 6k) + (25i + 10j - 15k)$$

$$35i + 14j - 21k = 35i + 14j - 21k$$

Q.5 Let $\vec{a} = 3i + j + 5k$ and $\vec{b} = -5i + 3j - 4k$ are two vectors and $m = 3$ and $n = 7$ are scalars, prove that:

$$(i) m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b} \quad (ii) (m + n)\vec{a} = m\vec{a} + n\vec{a}$$

SOLUTION:

$$(i) m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$$

$$3(3i + j + 5k - 5i + 3j - 4k) = 3\vec{a} + 3\vec{b}$$

$$3(3i + j + 5k - 5i + 3j - 4k)$$

$$= 3(3i + j + 5k) + 3(-5i + 3j - 4k)$$

$$3(-2i + 4j + k) = (9i + 3j + 15k) + (-15i + 9j - 12k)$$

$$-6i + 12j + 3k = -6i + 12j + 3k$$

$$(ii) (m + n)\vec{a} = m\vec{a} + n\vec{a}$$

$$(3 + 7)(3i + j + 5k) = 3(3i + j + 5k) + 7(3i + j + 5k)$$

$$10(3i + j + 5k) = (9i + 3j + 15k) + (21i + 7j + 35k)$$

$$30i + 10j + 50k = 30i + 10j + 50k$$

EXERCISE 3.4

Q.1 If $\vec{a} = 2i + 3j + 5k$, $\vec{b} = i - 2j - 3k$ and $\vec{c} = 5i + j - k$ then find:

(i) $\vec{a} \cdot \vec{b}$ (ii) $\vec{a} \cdot (\vec{b} + \vec{c})$ (iii) $(2\vec{a} + 3\vec{b}) \cdot (\vec{a} - 2\vec{b})$

SOLUTION:

(i) $\vec{a} \cdot \vec{b} = (2i + 3j + 5k) \cdot (i - 2j - 3k) = 2 - 6 - 15 = -19$

(ii) $\vec{a} \cdot (\vec{b} + \vec{c}) = (2i + 3j + 5k) \cdot (i - 2j - 3k + 5i + j - k)$
 $= (2i + 3j + 5k) \cdot (6i - j - 4k) = 12 - 3 - 20 = -11$

(iii) $(2\vec{a} + 3\vec{b}) \cdot (\vec{a} - 2\vec{b})$

$2\vec{a} + 3\vec{b} = 2(2i + 3j + 5k) + 3(i - 2j - 3k)$

$2\vec{a} + 3\vec{b} = 7i + 5k$

$\vec{a} - 2\vec{b} = (2i + 3j + 5k) - 2(i - 2j - 3k) = 7j + 11k$

$(2\vec{a} + 3\vec{b}) \cdot (\vec{a} - 2\vec{b}) = (7i + 5k) \cdot (7j + 11k) = 0 + 0 + 55$
 $= 55$

Q.2 If $\vec{a} = 5i - j - 3k$ and $\vec{b} = i + 3j - 5k$, then show that:

(i) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (ii) vectors $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular

SOLUTION:

(i) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

$(5i - j - 3k) \cdot (i + 3j - 5k) = (i + 3j - 5k) \cdot (5i - j - 3k)$

$5 - 3 + 15 = 5 - 3 + 15 \Rightarrow 17 = 17$

(ii) vectors $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular

$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b})$

$= (5i - j - 3k + i + 3j - 5k) \cdot \{(5i - j - 3k) - (i + 3j - 5k)\}$

$= (6i + 2j - 8k) \cdot (4i - 4j + 2k) = 24 - 8 - 16 = 0$

Vectors $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular

Q.3 Check whether the following vectors are orthogonal or not:

(i) $\vec{a} = 2i - 3j + 2k, \vec{b} = 5i + 8j + 7k$

(ii) $\vec{a} = i + 3j - 4k, \vec{b} = -i + j + 3k$

SOLUTION:

(i) $\vec{a} \cdot \vec{b} = (2i - 3j + 2k) \cdot (5i + 8j + 7k) = 10 - 24 + 14 = 0$

$\vec{a}$ and $\vec{b}$ are orthogonal

(ii) $\vec{a} = i + 3j - 4k, \vec{b} = -i + j + 3k$

$\vec{a} \cdot \vec{b} = (i + 3j - 4k) \cdot (-i + j + 3k) = -1 + 3 - 12 = -10$

$\vec{a}$ and $\vec{b}$ are not orthogonal

Q.4 Verify distributive laws for product for the following vectors:

(i) $\vec{a} = i - j + 2k, \vec{b} = 2i + 3j - k$ and $\vec{c} = 5i - j + 2k$

(ii) $\vec{a} = 3i - 5j + 7k, \vec{b} = -4i + j + k$ and $\vec{c} = 6i + j - 2k$

SOLUTION:

(i) $\vec{a} = i - j + 2k, \vec{b} = 2i + 3j - k$ and $\vec{c} = 5i - j + 2k$

$\vec{b} + \vec{c} = 2i + 3j - k + 5i - j + 2k = 7i + 2j + k$

$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

$(i - j + 2k) \cdot (7i + 2j + k)$

$= (i - j + 2k) \cdot (2i + 3j - k) + (i - j + 2k) \cdot (5i - j + 2k)$

$7 - 2 + 2 = 2 - 3 - 2 + 5 + 1 + 4$

$7 = 7$

(ii) $\vec{a} = 3i - 5j + 7k, \vec{b} = -4i + j + k$ and $\vec{c} = 6i + j - 2k$

A Help
Vectors $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular

Q.3 Check whether the following vectors are orthogonal or not:

(i) $\vec{a} = 2\vec{i} - 3\vec{j} + 2\vec{k}, \vec{b} = 5\vec{i} + 8\vec{j} + 7\vec{k}$

(ii) $\vec{a} = \vec{i} + 3\vec{j} - 4\vec{k}, \vec{b} = -\vec{i} + \vec{j} + 3\vec{k}$

SOLUTION:

(i) $\vec{a} \cdot \vec{b} = (2\vec{i} - 3\vec{j} + 2\vec{k}) \cdot (5\vec{i} + 8\vec{j} + 7\vec{k}) = 10 - 24 + 14 = 0$

$\vec{a}$ and $\vec{b}$ are orthogonal

(ii) $\vec{a} = \vec{i} + 3\vec{j} - 4\vec{k}, \vec{b} = -\vec{i} + \vec{j} + 3\vec{k}$

$\vec{a} \cdot \vec{b} = (\vec{i} + 3\vec{j} - 4\vec{k}) \cdot (-\vec{i} + \vec{j} + 3\vec{k}) = -1 + 3 - 12 = -10$

$\vec{a}$ and $\vec{b}$ are not orthogonal

Q.4 Verify distributive laws for product for the following vectors:

(i) $\vec{a} = \vec{i} - \vec{j} + 2\vec{k}, \vec{b} = 2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{c} = 5\vec{i} - \vec{j} + 2\vec{k}$

(ii) $\vec{a} = 3\vec{i} - 5\vec{j} + 7\vec{k}, \vec{b} = -4\vec{i} + \vec{j} + \vec{k}$ and $\vec{c} = 6\vec{i} + \vec{j} - 2\vec{k}$

SOLUTION:

(i) $\vec{a} = \vec{i} - \vec{j} + 2\vec{k}, \vec{b} = 2\vec{i} + 3\vec{j} - \vec{k}$ and $\vec{c} = 5\vec{i} - \vec{j} + 2\vec{k}$

$\vec{b} + \vec{c} = 2\vec{i} + 3\vec{j} - \vec{k} + 5\vec{i} - \vec{j} + 2\vec{k} = 7\vec{i} + 2\vec{j} + \vec{k}$

$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

$(\vec{i} - \vec{j} + 2\vec{k}) \cdot (7\vec{i} + 2\vec{j} + \vec{k})$

$= (\vec{i} - \vec{j} + 2\vec{k}) \cdot (2\vec{i} + 3\vec{j} - \vec{k}) + (\vec{i} - \vec{j} + 2\vec{k}) \cdot (5\vec{i} - \vec{j} + 2\vec{k})$

$7 - 2 + 2 = 2 - 3 - 2 + 5 + 1 + 4$

$7 = 7$

(ii) $\vec{a} = 3\vec{i} - 5\vec{j} + 7\vec{k}, \vec{b} = -4\vec{i} + \vec{j} + \vec{k}$ and $\vec{c} = 6\vec{i} + \vec{j} - 2\vec{k}$

$$\vec{b} + \vec{c} = -4i + j + k + 6i + j - 2k = 2i + 2j - k$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

$$(3i - 5j + 7k) \cdot (2i + 2j - k)$$

$$= (3i - 5j + 7k) \cdot (-4i + j + k) + (3i - 5j + 7k) \cdot (6i + j - 2k)$$

$$6 - 10 - 7 = -12 - 5 + 7 + 18 - 5 - 14$$

$$-11 = -11$$

Q.5 Find the angle between the given vectors:

(i) $i - 2j + 3k$ and $3i - 2j + k$

(ii) $3i + 5j - 3k$ and $i - 4j + k$

SOLUTION:

(i) $i - 2j + 3k$ and $3i - 2j + k$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{(i - 2j + 3k) \cdot (3i - 2j + k)}{\sqrt{(1)^2 + (-2)^2 + (3)^2} \sqrt{(3)^2 + (-2)^2 + (1)^2}} = \frac{3 + 4 + 3}{\sqrt{14} \sqrt{14}} = \frac{10}{14} = \frac{5}{7}$$

$$\theta = \cos^{-1} \left(\frac{5}{7} \right) = 44.41^\circ$$

(ii) $3i + 5j - 3k$ and $i - 4j + k$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{(3i + 5j - 3k) \cdot (i - 4j + k)}{\sqrt{(3)^2 + (5)^2 + (-3)^2} \sqrt{(1)^2 + (-4)^2 + (1)^2}} = \frac{3 - 20 - 3}{\sqrt{43} \sqrt{18}} = \frac{-20}{3\sqrt{86}}$$

$$\theta = \cos^{-1} \left(\frac{-20}{3\sqrt{86}} \right) = 135.96^\circ$$

Q.6 Find the direction cosines and direction angles for the following vectors.

(i) $2i + j - 3k$

(ii) $2i - 4j + 5k$

(iii) $-3i + 4j + 5k$

SOLUTION:

(i) $\vec{a} = 2i + j - 3k$

$$|\vec{a}| = \sqrt{(2)^2 + (1)^2 + (-3)^2} = \sqrt{14}$$

$$\cos \alpha = \frac{x}{|\vec{a}|} = \frac{2}{\sqrt{14}} \Rightarrow \alpha = \cos^{-1} \frac{2}{\sqrt{14}} = 57.68^\circ$$

$$\cos \beta = \frac{y}{|\vec{a}|} = \frac{1}{\sqrt{14}} \Rightarrow \beta = \cos^{-1} \frac{1}{\sqrt{14}} = 74.49^\circ$$

$$\cos \gamma = \frac{z}{|\vec{a}|} = \frac{-3}{\sqrt{14}} \Rightarrow \gamma = \cos^{-1} \frac{-3}{\sqrt{14}} = 143.30^\circ$$

(ii) $2\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$

$$|\vec{a}| = \sqrt{(2)^2 + (-4)^2 + (5)^2} = \sqrt{45} = 3\sqrt{5}$$

$$\cos \alpha = \frac{x}{|\vec{a}|} = \frac{2}{3\sqrt{5}} \Rightarrow \alpha = \cos^{-1} \frac{2}{3\sqrt{5}} = 72.65^\circ$$

$$\cos \beta = \frac{y}{|\vec{a}|} = -\frac{4}{3\sqrt{5}} \Rightarrow \beta = \cos^{-1} \left(-\frac{4}{3\sqrt{5}} \right) = 126.60^\circ$$

$$\cos \gamma = \frac{z}{|\vec{a}|} = \frac{5}{3\sqrt{5}} \Rightarrow \gamma = \cos^{-1} \frac{5}{3\sqrt{5}} = 41.81^\circ$$

(iii) $-3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}$

$$|\vec{a}| = \sqrt{(-3)^2 + (4)^2 + (5)^2} = \sqrt{50} = 5\sqrt{2}$$

$$\cos \alpha = \frac{x}{|\vec{a}|} = \frac{-3}{5\sqrt{2}} \Rightarrow \alpha = \cos^{-1} \left(\frac{-3}{5\sqrt{2}} \right) = 115.10^\circ$$

$$\cos \beta = \frac{y}{|\vec{a}|} = \frac{4}{5\sqrt{2}} \Rightarrow \beta = \cos^{-1} \frac{4}{5\sqrt{2}} = 55.55^\circ$$

$$\cos \gamma = \frac{z}{|\vec{a}|} = \frac{5}{5\sqrt{2}} \Rightarrow \gamma = \cos^{-1} \frac{1}{\sqrt{2}} = 45^\circ$$

Q.7 If α, β, γ are the direction angles of a vector then show that:

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

SOLUTION:

$$\vec{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$\cos \alpha = \frac{x}{|\vec{r}|} \Rightarrow \sin^2 \alpha = 1 - \frac{x^2}{x^2 + y^2 + z^2} = \frac{y^2 + z^2}{x^2 + y^2 + z^2}$$

$$\cos \beta = \frac{y}{|\vec{r}|} \Rightarrow \sin^2 \beta = 1 - \frac{y^2}{x^2+y^2+z^2} = \frac{x^2+z^2}{x^2+y^2+z^2}$$

$$\cos \gamma = \frac{z}{|\vec{r}|} \Rightarrow \sin^2 \gamma = 1 - \frac{z^2}{x^2+y^2+z^2} = \frac{x^2+y^2}{x^2+y^2+z^2}$$

$$\begin{aligned} \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma &= \frac{y^2+z^2}{x^2+y^2+z^2} + \frac{x^2+z^2}{x^2+y^2+z^2} + \frac{x^2+y^2}{x^2+y^2+z^2} \\ &= \frac{x^2+y^2+z^2+x^2+y^2+z^2}{x^2+y^2+z^2} = \frac{2(x^2+y^2+z^2)}{x^2+y^2+z^2} = 2 \end{aligned}$$

Q.8 If measures of two of the direction angles of a vector are 45° and 60° . Find the measure of third angle.

SOLUTION:

$$\alpha = 45^\circ$$

$$\beta = 60^\circ$$

$$(\cos 45^\circ)^2 + (\cos 60^\circ)^2 + \cos^2 \gamma = 1$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \cos^2 \gamma = 1$$

$$\frac{1}{2} + \frac{1}{4} + \cos^2 \gamma = 1$$

$$\cos^2 \gamma = 1 - \frac{3}{4} \Rightarrow \cos^2 \gamma = \frac{1}{4} \Rightarrow \cos \gamma = \pm \frac{1}{2}$$

$$\cos \gamma = \frac{1}{2} \Rightarrow \gamma = \cos^{-1} \frac{1}{2} \Rightarrow \gamma = 60^\circ$$

$$\cos \gamma = -\frac{1}{2} \Rightarrow \gamma = \cos^{-1} \left(-\frac{1}{2}\right) \Rightarrow \gamma = 120^\circ$$

Q.9 Find the work done by the force $\vec{F} = 7\vec{i} + 9\vec{j} - 11\vec{k}$ in moving an object along a straight line from $(4, 2, 7)$ to $(6, 4, 9)$.

SOLUTION:

$$\vec{d}_1 = (4, 2, 7)$$

$$\vec{d}_2 = (6, 4, 9)$$

$$\vec{d} = \vec{d}_2 - \vec{d}_1 = (6, 4, 9) - (4, 2, 7) = (6 - 4, 4 - 2, 9 - 7) = (2, 2, 2)$$

$$\vec{F} = 7\vec{i} + 9\vec{j} - 11\vec{k}$$

$$\begin{aligned}\text{Work done} &= \vec{F} \cdot \vec{d} = (7\vec{i} + 9\vec{j} - 11\vec{k}) \cdot (2\vec{i} + 2\vec{j} + 2\vec{k}) \\ &= 14 + 18 - 22 = 10 \text{ Joule}\end{aligned}$$

Q.10 Calculate the projection of $\vec{a}$ along $\vec{b}$ and projection of $\vec{b}$ along $\vec{a}$ when:

(i) $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$ and $\vec{b} = 3\vec{i} - \vec{j} - 5\vec{k}$

(ii) $\vec{a} = 2\vec{i} + 3\vec{j} + 2\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$

SOLUTION:

(i) $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$ and $\vec{b} = 3\vec{i} - \vec{j} - 5\vec{k}$

$$\text{Projection of } \vec{b} \text{ along } \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{(\vec{i} - 2\vec{j} + 3\vec{k}) \cdot (3\vec{i} - \vec{j} - 5\vec{k})}{\sqrt{(1)^2 + (-2)^2 + (3)^2}} = \frac{3 + 2 - 15}{\sqrt{14}} = \frac{-10}{\sqrt{14}}$$

$$\text{Projection of } \vec{a} \text{ along } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(\vec{i} - 2\vec{j} + 3\vec{k}) \cdot (3\vec{i} - \vec{j} - 5\vec{k})}{\sqrt{(3)^2 + (-1)^2 + (-5)^2}} = \frac{3 + 2 - 15}{\sqrt{35}} = \frac{-10}{\sqrt{35}}$$

(ii) $\vec{a} = 2\vec{i} + 3\vec{j} + 2\vec{k}$ and $\vec{b} = \vec{i} + 2\vec{j} + \vec{k}$

$$\text{Projection of } \vec{b} \text{ along } \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = \frac{(2\vec{i} + 3\vec{j} + 2\vec{k}) \cdot (\vec{i} + 2\vec{j} + \vec{k})}{\sqrt{(2)^2 + (3)^2 + (2)^2}} = \frac{2 + 6 + 2}{\sqrt{17}} = \frac{10}{\sqrt{17}}$$

$$\text{Projection of } \vec{a} \text{ along } \vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{(2\vec{i} + 3\vec{j} + 2\vec{k}) \cdot (\vec{i} + 2\vec{j} + \vec{k})}{\sqrt{(1)^2 + (2)^2 + (1)^2}} = \frac{2 + 6 + 2}{\sqrt{6}} = \frac{10}{\sqrt{6}}$$

Q.11 Calculate the work done by a force $\vec{F} = 3\vec{i} + 4\vec{j} + 5\vec{k}$ in displacing a body from position B to position A along a straight path. The position vectors of A and B respectively given as $\vec{r}_A = 2\vec{i} + 5\vec{j} - 2\vec{k}$ and $\vec{r}_B = 7\vec{i} + 3\vec{j} - 5\vec{k}$.

SOLUTION:

$$\begin{aligned}\vec{r} &= \vec{r}_A - \vec{r}_B = (2\vec{i} + 5\vec{j} - 2\vec{k}) - (7\vec{i} + 3\vec{j} - 5\vec{k}) \\ &= (-5\vec{i} + 2\vec{j} + 3\vec{k})\end{aligned}$$

$$W = \vec{F} \cdot \vec{r} = (3i + 4j + 5k) \cdot (-5i + 2j + 3k)$$

$$W = -15 + 8 + 15$$

$$W = 8$$

EXERCISE 3.5

Q.1 Find $\vec{a} \times \vec{b}$ and $\vec{b} \times \vec{a}$ if:

(i) $\vec{a} = 4i + j + k$ and $\vec{b} = i + 2j + 3k$

(ii) $\vec{a} = -2i + 5j - 3k$ and $\vec{b} = 3i + 2j + k$

SOLUTION:

(i) $\vec{a} = 4i + j + k$ and $\vec{b} = i + 2j + 3k$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 4 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = i \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} - j \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} + k \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(3 - 2) - j(12 - 1) + k(8 - 1)$$

$$\vec{a} \times \vec{b} = i - 11j + 7k$$

$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

$$\vec{b} \times \vec{a} = -(i - 11j + 7k) = -i + 11j - 7k$$

(ii) $\vec{a} = -2i + 5j - 3k$ and $\vec{b} = 3i + 2j + k$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ -2 & 5 & -3 \\ 3 & 2 & 1 \end{vmatrix} = i \begin{vmatrix} 5 & -3 \\ 2 & 1 \end{vmatrix} - j \begin{vmatrix} -2 & -3 \\ 3 & 1 \end{vmatrix} + k \begin{vmatrix} -2 & 5 \\ 3 & 2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(5 + 6) - j(-2 + 9) + k(-4 - 15)$$

$$\vec{a} \times \vec{b} = 11i - 7j - 19k$$

$$\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$$

$$\vec{b} \times \vec{a} = -(11i - 7j - 19k) = -11i + 7j + 19k$$

Q.2 Verify that $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$ for $\vec{a} = i - 7j + 7k$ and $\vec{b} = 3i - 2j + 2k$.

SOLUTION:

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 1 & -7 & 7 \\ 3 & -2 & 2 \end{vmatrix} = i \begin{vmatrix} -7 & 7 \\ -2 & 2 \end{vmatrix} - j \begin{vmatrix} 1 & 7 \\ 3 & 2 \end{vmatrix} + k \begin{vmatrix} 1 & -7 \\ 3 & -2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(-14 + 14) - j(2 - 21) + k(-2 + 21)$$

$$\vec{a} \times \vec{b} = 19j + 19k$$

$$\vec{b} \times \vec{a} = \begin{vmatrix} i & j & k \\ 3 & -2 & 2 \\ 1 & -7 & 7 \end{vmatrix} = i \begin{vmatrix} -2 & 2 \\ -7 & 7 \end{vmatrix} - j \begin{vmatrix} 3 & 2 \\ 1 & 7 \end{vmatrix} + k \begin{vmatrix} 3 & -2 \\ 1 & -7 \end{vmatrix}$$

$$\vec{b} \times \vec{a} = i(-14 + 14) - j(21 - 2) + k(-21 + 2)$$

$$\vec{b} \times \vec{a} = -19j - 19k$$

$$\vec{a} \times \vec{b} = -(19j + 19k)$$

$$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$$

Q.3 Find a unit vector which is orthogonal to both the vectors

$$\vec{a} = i - 2j + 3k \text{ and } \vec{b} = 3i - 2j + k$$

SOLUTION:

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 1 & -2 & 3 \\ 3 & -2 & 1 \end{vmatrix} = i \begin{vmatrix} -2 & 3 \\ -2 & 1 \end{vmatrix} - j \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + k \begin{vmatrix} 1 & -2 \\ 3 & -2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(-2 + 6) - j(1 - 9) + k(-2 + 6)$$

$$\vec{a} \times \vec{b} = 4i + 8j + 4k$$

$$|\vec{a} \times \vec{b}| = \sqrt{(4)^2 + (8)^2 + (4)^2} = \sqrt{96} = 4\sqrt{6}$$

$$\hat{u} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{4i + 8j + 4k}{4\sqrt{6}} = \frac{4(i + 2j + k)}{4\sqrt{6}} = \frac{i + 2j + k}{\sqrt{6}}$$

Q.4 Find a unit vector perpendicular to each of the vector $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ where $\vec{a} = 3i + 5j - k$ and $\vec{b} = i + 2j - 5k$.

SOLUTION:

$$\vec{a} + \vec{b} = (3i + 5j - k) + (i + 2j - 5k) = 4i + 7j - 6k$$

$$\vec{a} - \vec{b} = (3i + 5j - k) - (i + 2j - 5k) = 2i + 3j + 4k$$

$$\begin{aligned} (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) &= \begin{vmatrix} i & j & k \\ 4 & 7 & -6 \\ 2 & 3 & 4 \end{vmatrix} \\ &= i \begin{vmatrix} 7 & -6 \\ 3 & 4 \end{vmatrix} - j \begin{vmatrix} 4 & -6 \\ 2 & 4 \end{vmatrix} + k \begin{vmatrix} 4 & 7 \\ 2 & 3 \end{vmatrix} \end{aligned}$$

$$= i(28 + 18) - j(16 + 12) + k(12 - 14)$$

$$= 46i - 28j - 2k$$

$$\text{Magnitude} = \sqrt{(46)^2 + (-28)^2 + (-2)^2} = \sqrt{2904} = 2\sqrt{726}$$

$$\hat{u} = \frac{2(23i - 14j - k)}{2\sqrt{726}} = \frac{(23i - 14j - k)}{\sqrt{726}}$$

Q.5 Find the area of the parallelogram whose adjacent sides are represented by $\vec{a} = 5i - j + 3k$ and $\vec{b} = i + k$.

SOLUTION:

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 5 & -1 & 3 \\ 1 & 0 & 1 \end{vmatrix} = i \begin{vmatrix} -1 & 3 \\ 0 & 1 \end{vmatrix} - j \begin{vmatrix} 5 & 3 \\ 1 & 1 \end{vmatrix} + k \begin{vmatrix} 5 & -1 \\ 1 & 0 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(-1 - 0) - j(5 - 3) + k(0 + 1)$$

$$\vec{a} \times \vec{b} = -i + 2j + k$$

$$|\vec{a} \times \vec{b}| = \sqrt{(-1)^2 + (2)^2 + (1)^2} = \sqrt{6}$$

$$\text{Area of parallelogram} = |\vec{a} \times \vec{b}| = \sqrt{6} \text{ square units}$$

Q.6 Find the area of triangle with vertices $A(1, 1, 2)$, $B(2, 3, 5)$ and $C(1, 5, 5)$.

SOLUTION:

$$\vec{AB} = B - A = (2, 3, 5) - (1, 1, 2) = (2 - 1, 3 - 1, 5 - 2) = (1, 2, 3)$$

$$\vec{AC} = C - A = (1, 5, 5) - (1, 1, 2) = (1 - 1, 5 - 1, 5 - 2) = (0, 4, 3)$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 0 & 4 & 3 \end{vmatrix} = i \begin{vmatrix} 2 & 3 \\ 4 & 3 \end{vmatrix} - j \begin{vmatrix} 1 & 3 \\ 0 & 3 \end{vmatrix} + k \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix}$$

$$\vec{AB} \times \vec{AC} = i(6 - 12) - j(3 - 0) + k(4 - 0)$$

$$\vec{AB} \times \vec{AC} = -6i - 3j + 4k$$

$$|\vec{AB} \times \vec{AC}| = \sqrt{(-6)^2 + (-3)^2 + (4)^2} = \sqrt{61}$$

$$\text{Area of triangle} = \frac{|\vec{AB} \times \vec{AC}|}{2} = \frac{\sqrt{61}}{2} \text{ square units}$$

Q.7 If $\vec{a} = 2i + j + 7k$, $\vec{b} = i - 3j + 5k$ and $\vec{c} = -i + 5j + k$ then verify that $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$.

SOLUTION:

$$\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

$$\vec{b} + \vec{c} = i - 3j + 5k - i + 5j + k = 2j + 6k$$

$$\vec{a} \times (\vec{b} + \vec{c}) = \begin{vmatrix} i & j & k \\ 2 & 1 & 7 \\ 0 & 2 & 6 \end{vmatrix} = i \begin{vmatrix} 1 & 7 \\ 2 & 6 \end{vmatrix} - j \begin{vmatrix} 2 & 7 \\ 0 & 6 \end{vmatrix} + k \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix}$$

$$\vec{a} \times (\vec{b} + \vec{c}) = i(6 - 14) - j(12 - 0) + k(4 - 0)$$

$$\vec{a} \times (\vec{b} + \vec{c}) = -8i - 12j + 4k$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 2 & 1 & 7 \\ 1 & -3 & 5 \end{vmatrix} = i \begin{vmatrix} 1 & 7 \\ -3 & 5 \end{vmatrix} - j \begin{vmatrix} 2 & 7 \\ 1 & 5 \end{vmatrix} + k \begin{vmatrix} 2 & 1 \\ 1 & -3 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(5 + 21) - j(10 - 7) + k(-6 - 1)$$

$$\vec{a} \times \vec{b} = 26i - 3j - 7k$$

$$\vec{a} \times \vec{c} = \begin{vmatrix} i & j & k \\ 2 & 1 & 7 \\ -1 & 5 & 1 \end{vmatrix} = i \begin{vmatrix} 1 & 7 \\ 5 & 1 \end{vmatrix} - j \begin{vmatrix} 2 & 7 \\ -1 & 1 \end{vmatrix} + k \begin{vmatrix} 2 & 1 \\ -1 & 5 \end{vmatrix}$$

$$\vec{a} \times \vec{c} = i(1 - 35) - j(2 + 7) + k(10 + 1)$$

$$\vec{a} \times \vec{c} = -34i - 9j + 11k$$

$$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} = 26i - 3j - 7k - 34i - 9j + 11k$$

$$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} = -8i - 12j + 4k$$

Q.8 Using cross product, find the angles between $i - 7j + 7k$ and $3i - 2j + 2k$.

SOLUTION:

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 1 & -7 & 7 \\ 3 & -2 & 2 \end{vmatrix} = i \begin{vmatrix} -7 & 7 \\ -2 & 2 \end{vmatrix} - j \begin{vmatrix} 1 & 7 \\ 3 & 2 \end{vmatrix} + k \begin{vmatrix} 1 & -7 \\ 3 & -2 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = i(-14 + 14) - j(2 - 21) + k(-2 + 21)$$

$$\vec{a} \times \vec{b} = 19j + 19k$$

$$|\vec{a} \times \vec{b}| = \sqrt{(19)^2 + (19)^2} = \sqrt{2(19)^2} = 19\sqrt{2}$$

$$|\vec{a}| = \sqrt{(1)^2 + (-7)^2 + (7)^2} = \sqrt{99} = 3\sqrt{11}$$

$$|\vec{b}| = \sqrt{(3)^2 + (-2)^2 + (2)^2} = \sqrt{17}$$

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} = \frac{19\sqrt{2}}{(3\sqrt{11})(\sqrt{17})} \Rightarrow \theta = \sin^{-1} \left(\frac{19\sqrt{2}}{3\sqrt{187}} \right) = 40.91^\circ$$

Q.9 Find the moment about a point $A(1, 3, 5)$ of the force $\vec{F} = i + 2j - k$ applied at $B(3, -2, 5)$.

SOLUTION:

$$\vec{r} = B - A = (3, -2, 5) - (1, 3, 5) = (3 - 1, -2 - 3, 5 - 5)$$

$$\vec{r} = (2, -5, 0) = 2i - 5j$$

$$\vec{F} = i + 2j - k$$

$$\vec{r} \times \vec{F} = \begin{vmatrix} i & j & k \\ 2 & -5 & 0 \\ 1 & 2 & -1 \end{vmatrix} = i \begin{vmatrix} -5 & 0 \\ 2 & -1 \end{vmatrix} - j \begin{vmatrix} 2 & 0 \\ 1 & -1 \end{vmatrix} + k \begin{vmatrix} 2 & -5 \\ 1 & 2 \end{vmatrix}$$

$$\vec{r} \times \vec{F} = i(5 - 0) - j(-2 - 0) + k(4 + 5)$$

$$\vec{r} \times \vec{F} = 5i + 2j + 9k$$

A Help Desk
 Q.10 If you apply 40N force to the end of 0.3 meter wrench at an angle of 110° . Calculate the magnitude of the moment $\vec{M}_0$ produced by applied force.

SOLUTION:

A force of 40N acting at an angle of 110° at a distance 0.3 m from it. Therefore magnitude of the moment produced by applied force is

$$\tau = 40 \sin 110^\circ (0.3) = 11.27 \text{ N}$$

EXERCISE 3.6

Q.1 Find the scalar tripe product of vectors $i + 2j + 3k$, $-i - j + k$ and $i + j + k$.

SOLUTION:

$$\begin{vmatrix} 1 & 2 & 3 \\ -1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} - 2 \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} + 3 \begin{vmatrix} -1 & -1 \\ 1 & 1 \end{vmatrix}$$

$$= 1(-1 - 1) - 2(-1 - 1) + 3(-1 + 1)$$

$$= -2 + 4 + 0$$

$$= 2$$

Q.2 Compute $[i + j, i, i - j + k]$.

SOLUTION:

$$\begin{vmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & -1 & 1 \end{vmatrix}$$

$$= 1 \begin{vmatrix} 0 & 0 \\ -1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} + 0$$

$$= 1(0 - 0) - 1(1 - 0)$$

$$= 0 - 1$$

$$= -1$$

Q.3 Find the volume of parallelopiped whose edges are represented by the vectors:

(i) $3i + 2j - k; \quad i - 2j + k; \quad i + 2j - 4k$

(ii) $i - 2j + 3k; \quad 2i - j - k; \quad i + 2j - 4k$

SOLUTION:

$$(i) \begin{vmatrix} 3 & 2 & -1 \\ 1 & -2 & 1 \\ 1 & 2 & -4 \end{vmatrix}$$

$$= 3 \begin{vmatrix} -2 & 1 \\ 2 & -4 \end{vmatrix} - 2 \begin{vmatrix} 1 & 1 \\ 1 & -4 \end{vmatrix} - 1 \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix}$$

$$= 3(8 - 2) - 2(-4 - 1) - 1(2 + 2)$$

$$= 3(6) - 2(-5) - 1(4)$$

$$= 24$$

(ii) $i - 2j + 3k; \quad 2i - j - k; \quad i + 2j - 4k$

$$= \begin{vmatrix} 1 & -2 & 3 \\ 2 & -1 & -1 \\ 1 & 2 & -4 \end{vmatrix}$$

$$= 1 \begin{vmatrix} -1 & -1 \\ 2 & -4 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ 1 & -4 \end{vmatrix} + 3 \begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix}$$

$$= 1(4 + 2) + 2(-8 + 1) + 3(4 + 1)$$

$$= 7$$

Q.4 Find the volume of the tetrahedron whose vertices are

$A(2,1,8), B(3,2,9), C(2,1,4)$ and $D(3,3,10)$

SOLUTION:

$$\text{Volume} = \frac{1}{6} [\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}]$$

$$\overrightarrow{AB} = B - A = (3,2,9) - (2,1,8) = (1,1,1)$$

$$\overrightarrow{AC} = C - A = (2, 1, 4) - (2, 1, 8) = (0, 0, -4)$$

$$\overrightarrow{AD} = D - A = (3, 3, 10) - (2, 1, 8) = (1, 2, 2)$$

$$6V = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 1 & 2 & 2 \end{vmatrix}$$

$$6V = 1 \begin{vmatrix} 0 & -4 \\ 2 & 2 \end{vmatrix} - 1 \begin{vmatrix} 0 & -4 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 0 & 0 \\ 1 & 2 \end{vmatrix}$$

$$6V = 1(0 + 8) - 1(0 + 4) + 1(0 - 0)$$

$$6V = 8 - 4$$

$$6V = 4$$

$$V = \frac{4}{6} = \frac{2}{3} \text{ cubic unit}$$

Q.5 Show that the vectors $4i - j + k$, $3i - 2j - k$ and $i + j + 2k$ are coplanar.

SOLUTION:

$$\text{Volume} = \begin{vmatrix} 4 & -1 & 1 \\ 3 & -2 & -1 \\ 1 & 1 & 2 \end{vmatrix}$$

$$= 4 \begin{vmatrix} -2 & -1 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 3 & -1 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix}$$

$$= 4(-4 + 1) + 1(6 + 1) + 1(3 + 2)$$

$$= 0$$

Hence vectors are coplanar

Q.6 Find λ if the vectors $i + j + 2k$, $\lambda i - j + k$ and $3i - 2j - k$ are coplanar.

SOLUTION:

$$\text{Volume} = 0$$

$$\begin{vmatrix} 1 & 1 & 2 \\ \lambda & -1 & 1 \\ 3 & -2 & -1 \end{vmatrix} = 0$$

$$1 \begin{vmatrix} -1 & 1 \\ -2 & -1 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 2 \\ -2 & -1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = 0$$

$$1(1+2) - \lambda(-1+4) + 3(1+2) = 0$$

$$3 - 3\lambda + 9 = 0$$

$$-3\lambda = -12$$

$$\lambda = \frac{12}{3} \Rightarrow \boxed{\lambda = 4}$$

Q.7 Show that $i + j + k$, $i - 2j + 3k$ and $2i - j + 3k$ are not coplanar vectors.

SOLUTION:

$$\begin{aligned} \text{Volume} &= \begin{vmatrix} 1 & 1 & 1 \\ 1 & -2 & 3 \\ 2 & -1 & 3 \end{vmatrix} \\ &= 1 \begin{vmatrix} -2 & 3 \\ -1 & 3 \end{vmatrix} - 1 \begin{vmatrix} 1 & 3 \\ 2 & 3 \end{vmatrix} + 1 \begin{vmatrix} 1 & -2 \\ 2 & -1 \end{vmatrix} \\ &= 1(-6+3) - 1(3-6) + 1(-1+4) \\ &= -3 - 1(-3) + 1(3) \\ &= 3 \end{aligned}$$

Hence vectors are not coplanar.

REVIEW EXERCISE 3

1. Multiple Choice Question.

(i) The unit vector in the direction of $\vec{r} = 2i + j - k$ is:

(a) $\frac{1}{\sqrt{6}}(2i + j - k)$

(b) $\sqrt{6}(2i + j - k)$

(c) $6(2i + j - k)$

(d) $\frac{1}{6}(2i + j - k)$

(ii) The magnitude of $\vec{a} \times \vec{b}$ represents the _____ of a parallelogram with adjacent sides $\vec{a}$ and $\vec{b}$.

(a) opposite sides

(b) diagonal

(c) area

(d) volume

(iii) The volume of tetrahedron determined by vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is:

(a) $[\vec{a}, \vec{b}, \vec{c}]$

(b) $\frac{1}{3}[\vec{a}, \vec{b}, \vec{c}]$

(c) $\frac{1}{6}[\vec{a}, \vec{b}, \vec{c}]$

(d) none

(iv) $\vec{a} \cdot (\vec{b} \times \vec{c}) =$ _____

(a) $\vec{a} \cdot \vec{b} \cdot \vec{c}$

(b) $\vec{a} \times \vec{b} \times \vec{c}$

(c) $\vec{b} \cdot (\vec{a} \times \vec{c})$

(d) $(\vec{a} \times \vec{b}) \cdot \vec{c}$

(v) If $\vec{a}$ and $\vec{b}$ are parallel then $\vec{a} \cdot \vec{b} =$ _____.

(a) 1

(b) -1

(c) 0

(d) ab

(vi) If three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar, then

$[\vec{a}, \vec{b}, \vec{c}] =$ _____

(a) 1

(b) 0

(c) $\vec{c}$

(d) $\vec{a}$

(vii) Direction cosines of the vector $i + j + k$ are:

(a) $\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

(b) $\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

(c) 1, 1, -1

(d) $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$

(viii) $\vec{a} \times \vec{b}$ is _____ to plane of $\vec{a}$ and $\vec{b}$:

(a) parallel

(b) perpendicular

(c) opposite

(d) none of these

(ix) If $\vec{a} = \overrightarrow{P_1P_2}$, where $P_1(0, 0, 1)$ and $P_2(-3, 1, 2)$, then $|\vec{a}| =$

(a) $\sqrt{12}$

(b) $\sqrt{10}$

(c) $\sqrt{13}$

(d) $\sqrt{11}$

(x) If $\vec{p} = 4\vec{i} + 6\vec{k}$ and $\vec{q} = 6\vec{i} - 4\vec{j}$ then $|\vec{p} - \vec{q}|$ is

(a) $\sqrt{8}$

(b) $2\sqrt{14}$

(c) $2\sqrt{26}$

(d) $2\sqrt{3}$

(xi) For non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a} \times \vec{b}$ is a unit vector and $|\vec{a}| = |\vec{b}| = \sqrt{2}$, then angle between vectors $\vec{a}$ and $\vec{b}$ is:

(a) $\frac{\pi}{2}$

(b) $\frac{\pi}{3}$

(c) $\frac{\pi}{6}$

(d) $-\frac{\pi}{2}$

(xii) Magnitude of a vector $\vec{a} = 3\vec{i} - \vec{j} + 2\vec{k}$ is:

(a) 13

(b) $\sqrt{12}$

(c) $\sqrt{14}$

(d) $\sqrt{11}$

(xiii) The position vector of the point $(1, 0, 2)$ is:

(a) $\vec{i} + \vec{j} + 2\vec{k}$

(b) $\vec{i} + 2\vec{j}$

(c) $2\vec{i} + 3\vec{k}$

(d) $\vec{i} + 2\vec{k}$

(xiv) If O be the origin and $\overrightarrow{OP} = 2\vec{i} + 3\vec{j} - 4\vec{k}$ and $\overrightarrow{OQ} = 5\vec{i} + 4\vec{j} - 3\vec{k}$, then $\overrightarrow{PQ}$ is equal to:

(a) $7\vec{i} + 7\vec{j} - 7\vec{k}$

(b) $-3\vec{i} + \vec{j} - \vec{k}$

(c) $3\vec{i} + \vec{j} + \vec{k}$

(d) $-7\vec{i} - 7\vec{j} + 7\vec{k}$

(xv) $k \times j = \underline{\hspace{2cm}}$.

- (a) i (b) k
(c) j (d) $-i$

(xvi) If $|\vec{a} \times \vec{b}| = |\vec{a} \cdot \vec{b}|$ then the angle between $\vec{a}$ and $\vec{b}$:

- (a) 0 (b) $\frac{\pi}{2}$
(c) $\frac{\pi}{4}$ (d) π

(xvii) The distance of the point $(-3, 4, 5)$ from the origin is:

- (a) 50 (b) $5\sqrt{2}$
(c) 6 (d) none of these

(xviii) The vector having, initial and terminal points as $(2, 5, 0)$ and $(-3, 7, 4)$ respectively is:

- (a) $-i + 12j + 4k$ (b) $-5i + 3j - 4k$
(c) $-5i + 2j + 4k$ (d) $-i + j + k$

(xix) If $|\vec{a}| = 10$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = 0$, then the value of $|\vec{a} \times \vec{b}|$ is:

- (a) 10 (b) 16
(c) 14 (d) 20

(xx) The number of vectors of unit length perpendicular to the plane of the vectors $\vec{a} = 2i + j + 2k$ and $\vec{a} = j + k$ is:

- (a) one (b) two
(c) three (d) infinite

Q.2 Write $\vec{PQ}$ in the form $xi + yj$

$$(ii) P\left(-\frac{5}{2}, 9\right), Q(10, 8)$$

$$(i) P(4, 6), Q(-4, 5)$$

SOLUTION:

$$(i) \overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (-4, 5) - (4, 6) = (-4 - 4, 5 - 6)$$

$$\overrightarrow{PQ} = (-8, -1) = -8i - j$$

$$(ii) \overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (10, 8) - \left(-\frac{5}{2}, 9\right) = \left(10 + \frac{5}{2}, 8 - 9\right)$$

$$\overrightarrow{PQ} = \left(\frac{25}{2}, -1\right) = \frac{25}{2}i - j$$

Q.3 Find the vector PQ joining the points P(3, 4, 2) and

Q(-2, 3, 4) and find the direction cosines of $\overrightarrow{PQ}$.

SOLUTION:

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (-2, 3, 4) - (3, 4, 2) = (-2 - 3, 3 - 4, 4 - 2)$$

$$\overrightarrow{PQ} = (-5, -1, 2)$$

$$\cos \alpha = \frac{-5}{\sqrt{(-5)^2 + (-1)^2 + (2)^2}} = \frac{-5}{\sqrt{30}}$$

$$\cos \beta = \frac{-1}{\sqrt{(-5)^2 + (-1)^2 + (2)^2}} = \frac{-1}{\sqrt{30}}$$

$$\cos \gamma = \frac{2}{\sqrt{(-5)^2 + (-1)^2 + (2)^2}} = \frac{2}{\sqrt{30}}$$

Q.4 Show that P(2, 3, 6), Q(3, 7, 4) and R(4, 11, 2) are collinear.

SOLUTION:

$$\text{Volume} = \begin{vmatrix} 2 & 3 & 6 \\ 3 & 7 & 4 \\ 4 & 11 & 2 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 7 & 4 \\ 11 & 2 \end{vmatrix} - 3 \begin{vmatrix} 3 & 4 \\ 4 & 2 \end{vmatrix} + 6 \begin{vmatrix} 3 & 7 \\ 4 & 11 \end{vmatrix}$$

$$= 2(14 - 44) - 3(6 - 16) + 6(33 - 28)$$

$$= 0$$

Hence vectors are collinear

Q.5 Show that the vectors $\frac{1}{7}(i + j + k)$, $2i - 3j + k$ and $4i + j - 5k$ are mutually perpendicular.

SOLUTION:

For perpendicular vectors $\vec{a} \cdot (\vec{b} + \vec{c}) = 0$

$$\vec{a} = \frac{1}{7}(i + j + k)$$

$$\vec{b} + \vec{c} = (2i - 3j + k) + (4i + j - 5k)$$

$$\vec{b} + \vec{c} = 6i - 2j - 4k$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \frac{1}{7}(i + j + k) \cdot (6i - 2j - 4k)$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \frac{1}{7}(6 - 2 - 4)$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \frac{1}{7}(0)$$

$$\vec{a} \cdot (\vec{b} + \vec{c}) = 0$$

Q.6 Find the area of parallelogram whose adjacent sides are $3i - 5j + 6k$ and $i + 3j - 4k$. Also find unit vector parallel to its diagonal through common initial point.

SOLUTION:

$$\vec{a} = 3i - 5j + 6k, \vec{b} = i + 3j - 4k$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 3 & -5 & 6 \\ 1 & 3 & -4 \end{vmatrix}$$

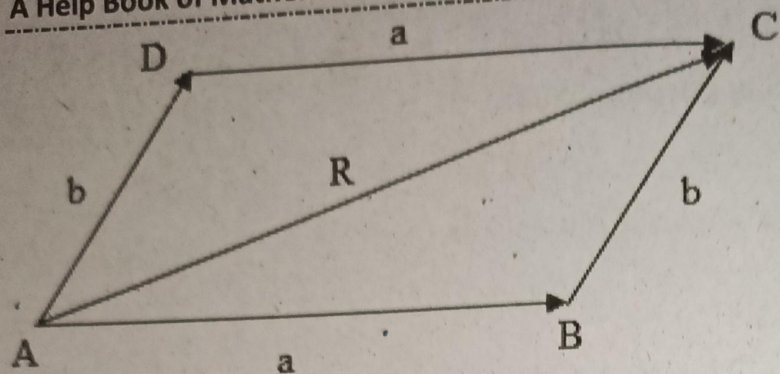
$$= i \begin{vmatrix} -5 & 6 \\ 3 & -4 \end{vmatrix} - j \begin{vmatrix} 3 & 6 \\ 1 & -4 \end{vmatrix} + k \begin{vmatrix} 3 & -5 \\ 1 & 3 \end{vmatrix}$$

$$= i(20 - 18) - j(-12 - 6) + k(9 + 5)$$

$$= 2i + 18j + 14k$$

$$|\vec{a} \times \vec{b}| = \sqrt{(2)^2 + (18)^2 + (14)^2} = \sqrt{524} = 2\sqrt{131}$$

$$\text{Area of parallelogram} = |\vec{a} \times \vec{b}| = 2\sqrt{131} \text{ square units}$$



$$\vec{R} = \vec{a} + \vec{b}$$

$$= 3i - 5j + 6k + i + 3j - 4k$$

$$= 4i - 2j + 2k$$

$$\text{Unit vector } \hat{u} = \frac{\vec{R}}{|\vec{R}|} = \frac{4i - 2j + 2k}{\sqrt{24}} = \frac{2(2i - j + k)}{2\sqrt{6}} = \frac{(2i - j + k)}{\sqrt{6}}$$

Q.7 Prove that $[\vec{p} + \vec{q}, \vec{q} + \vec{r}, \vec{r} + \vec{p}] = 2[\vec{p}, \vec{q}, \vec{r}]$.

SOLUTION:

$$[\vec{p} + \vec{q}, \vec{q} + \vec{r}, \vec{r} + \vec{p}]$$

$$= [\vec{p} + \vec{q} + 0\vec{r}, 0\vec{p} + \vec{q} + \vec{r}, \vec{p} + 0\vec{q} + \vec{r}]$$

$$= \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} [\vec{p}, \vec{q}, \vec{r}]$$

$$= \left\{ 1 \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + 0 \right\} [\vec{p}, \vec{q}, \vec{r}]$$

$$= \{1(1 - 0) - (0 - 1)\} [\vec{p}, \vec{q}, \vec{r}]$$

$$= \{1 + 1\} [\vec{p}, \vec{q}, \vec{r}]$$

$$= 2[\vec{p}, \vec{q}, \vec{r}]$$

Q.8 For the vectors $\vec{a} = i + 2j + k$, $\vec{b} = 3i - j - k$ and $\vec{c} = i + 8j + k$, verify the distributive property of cross product.

SOLUTION:

$$\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$$

$$\vec{b} + \vec{c} = (3i - j - k) + (i + 8j + k) = 4i + 7j$$

$$\vec{a} \times (\vec{b} + \vec{c}) = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 4 & 7 & 0 \end{vmatrix}$$

$$= i \begin{vmatrix} 2 & 1 \\ 7 & 0 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 4 & 0 \end{vmatrix} + k \begin{vmatrix} 1 & 2 \\ 4 & 7 \end{vmatrix}$$

$$= i(0 - 7) - j(0 - 4) + k(7 - 8)$$

$$\boxed{\vec{a} \times (\vec{b} + \vec{c}) = -7i + 4j - k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 3 & -1 & -1 \end{vmatrix}$$

$$= i \begin{vmatrix} 2 & 1 \\ -1 & -1 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} + k \begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix}$$

$$= i(-2 + 1) - j(-1 - 3) + k(-1 - 6)$$

$$\vec{a} \times \vec{b} = -i + 4j - 7k$$

$$\vec{a} \times \vec{c} = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ 1 & 8 & 1 \end{vmatrix}$$

$$= i \begin{vmatrix} 2 & 1 \\ 8 & 1 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} + k \begin{vmatrix} 1 & 2 \\ 1 & 8 \end{vmatrix}$$

$$= i(2 - 8) - j(1 - 1) + k(8 - 2)$$

$$\vec{a} \times \vec{c} = -6i + 6k$$

$$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} = -i + 4j - 7k - 6i + 6k$$

$$\vec{a} \times \vec{b} + \vec{a} \times \vec{c} = -7i + 4j - k$$